TECHNICAL DOCUMENT

AI-Optimized Leakage Test Time Reduction

Cutting IDDQ, standby-leakage and pin-leakage test time on ATE with settling optimization, early-termination prediction, vector reduction and adaptive test

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Scope	Wafer sort and final test of digital and mixed-signal ICs; IDDQ, standby/mode leakage, pin leakage
Audience	Test engineers, product engineers, test-data/ML engineers, quality engineers

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1. Executive summary

Leakage tests—IDDQ, standby and power-mode leakage, and pin leakage—are DC measurements. Each one needs the device and the tester's measurement path to settle before the reading is valid, and the measurement is repeated across many vectors, power modes, pins and temperatures. As a result, leakage tests often take a disproportionate share of total test time even though they are electrically simple.

Most of that time is **waiting**, and most of the waiting is set by worst-case assumptions: one fixed settling delay for every vector, every vector for every part, every pin measured individually, and a full hot insertion for every part. This document describes how to remove that conservatism safely:

- **Settling-time characterization and adaptive settling**: replace fixed worst-case waits with measured, per-vector waits or stability-based stopping. No AI required; usually the largest single gain.
- **Early-termination prediction**: fit or learn the settling transient from the first few samples and decide pass/fail without waiting for full decay, with guard bands that send near-limit parts to the full test.
- **IDDQ vector reduction**: select the small subset of vectors that carries almost all of the screening information, and predict the rest.
- **Adaptive test**: use upstream data (wafer sort, process monitors, wafer maps) to route low-risk parts through a reduced leakage flow and high-risk parts through the full flow.
- **Temperature-insertion reduction**: predict hot leakage from room/cold measurements and process monitors.
- **Group pin-leakage testing** with bisection on failure, plus multi-site and flow-order improvements.

Combined, these methods commonly reduce leakage test time by 30–70%, and more when part of a temperature insertion can be removed. The constraint is quality: every method is paired with guard bands, shadow-mode validation, audit sampling and drift monitoring so that test escapes and overkill do not increase.

> **Note on numbers.** Time and savings figures in this document are typical or illustrative values. Actual results depend on the product, tester, load board, current program and quality requirements. Always measure against your own baseline.

2. Leakage tests on ATE

Test	Purpose	Typical method	Main time drivers
IDDQ (quiescent supply current)	Detect bridging, gate-oxide and other defects that raise static current	Apply scan vector, stop clocks, wait, measure V_{DD} current with DPS/PMU; repeat for 10–20+ vectors	Settling per vector; vector count; ranging
Standby / power-mode leakage (ISB)	Verify leakage in sleep, retention and off modes meets datasheet	Configure mode via scan/JTAG/registers, wait for regulators and rails, measure per supply	Mode-entry sequences; regulator settling; multiple supplies
Pin leakage (I_{IL}/I_{IH}, I_{OZ})	Detect input/tri-state leakage from ESD damage or defects	Force voltage per pin with PMU, measure nA–µA current	Pin count; per-pin PMU sequencing; low-current ranging
Multi-temperature leakage	Guarantee limits at hot (and cold) corners; screen temperature-activated defects	Repeat leakage tests at separate temperature insertions	Entire extra insertion; handler soak time

Table 1. Leakage tests and their time drivers.

3. Anatomy of leakage test time

3.1 Time model

For one part, leakage test time can be written as

$$T_{leak} = \Sigma_{vectors}\,(t_{apply} + t_{settle} + t_{meas}) + \Sigma_{modes}\,(t_{entry} + t_{settle} + t_{meas}) + \Sigma_{pins}\,t_{pin} + T_{hot}$$

In most programs t_{settle} dominates the IDDQ and mode terms, and the pin term grows linearly with pin count.

3.2 Why settling is slow

- **Decoupling and rail capacitance.** On-die, package and load-board capacitance must discharge from the dynamic current of the last vector to the quiescent level. The decay is roughly exponential, $\tau \approx R_{eff} \cdot C_{total}$, and several time constants are needed for a µA-level reading.
- **On-chip regulators and power switches.** LDOs, power-gating switches and retention rails have their own settling and may show slow tails.
- **Tester measurement path.** Device power supply (DPS) or PMU range changes, filtering and integration time add delay; low-current ranges are slower.
- **Noise averaging.** Measurement noise falls as $\sigma \propto 1/\sqrt{N}$ with N samples, so tight limits push longer integration.
- **Worst-case programming.** Waits are typically set once, for the slowest vector on the slowest part at bring-up, then never revisited.

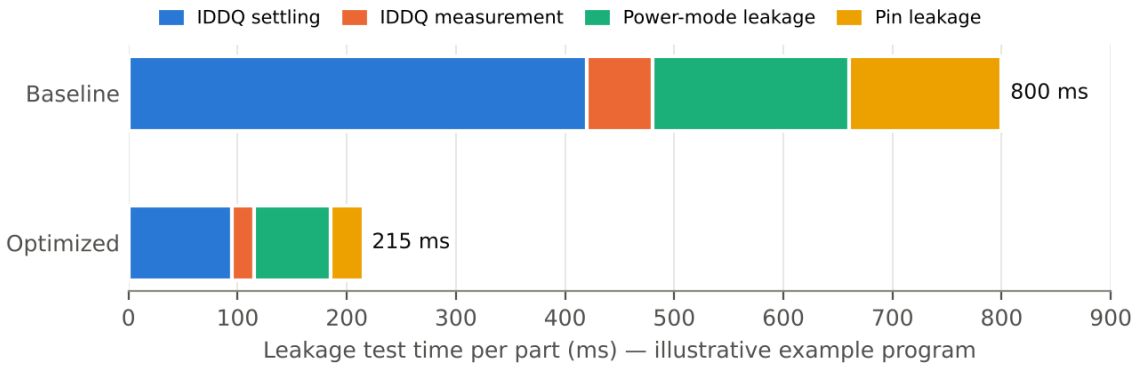


Figure 1. Example breakdown of leakage test time before and after optimization (illustrative program). Settling is the largest component and the largest opportunity.

4. Baseline (non-AI) optimizations

These should be done first: they are low-risk, they often deliver most of the saving, and they produce the characterization data that the AI methods need.

Technique	How	Typical effect	Risk / caveat
Per-vector settling characterization	Capture full settling curves for every vector, mode and corner on a characterization lot; set each wait to worst case + margin	Large reduction in IDDQ settling time	Must cover process, temperature and site-to-site variation
Adaptive (stability-based) settling	Sample repeatedly; stop when $\|\Delta I\|$ between samples < threshold for k consecutive samples	Further savings on fast-settling parts	Needs fast sampling; protect against slow tails
Fixed ranging	Pre-set the measurement range per test to avoid autorange delays	Removes range-change time	Range must cover limit with resolution
Integration-time tuning	Use the shortest integration that keeps measurement noise well below the guard band	Shorter t_{meas}	Re-check gauge repeatability (GR&R)
Load-board design	Minimize board decap on leakage-tested rails or add switchable/active discharge	Shorter τ	Must not compromise functional/at-speed tests
Multi-site and parallel measurement	Measure all sites simultaneously; avoid serialized PMU use	Lower time per part	Resource limits on DPS/PMU channels
Group pin leakage	Measure groups of pins together; bisect only failing groups (section 10)	Large pin-leakage reduction	Group leakage must stay resolvable vs. limit
Fail-fast ordering	Run tests most likely to fail first	Lower average time on failing parts	Little effect when yield is high

Table 2. Baseline optimizations.

5. Early-termination prediction

5.1 Principle

After a vector is applied and clocks stop, the supply current decays towards its quiescent value:

$$I(t) = I_{final} + A \cdot e^{-t/\tau}$$ *(often a sum of two exponentials for board and on-die effects)*

Instead of waiting for the transient to vanish, sample it during the first part of the decay, estimate I_{final}, and make the pass/fail decision immediately when the estimate is clearly inside or outside the limit.

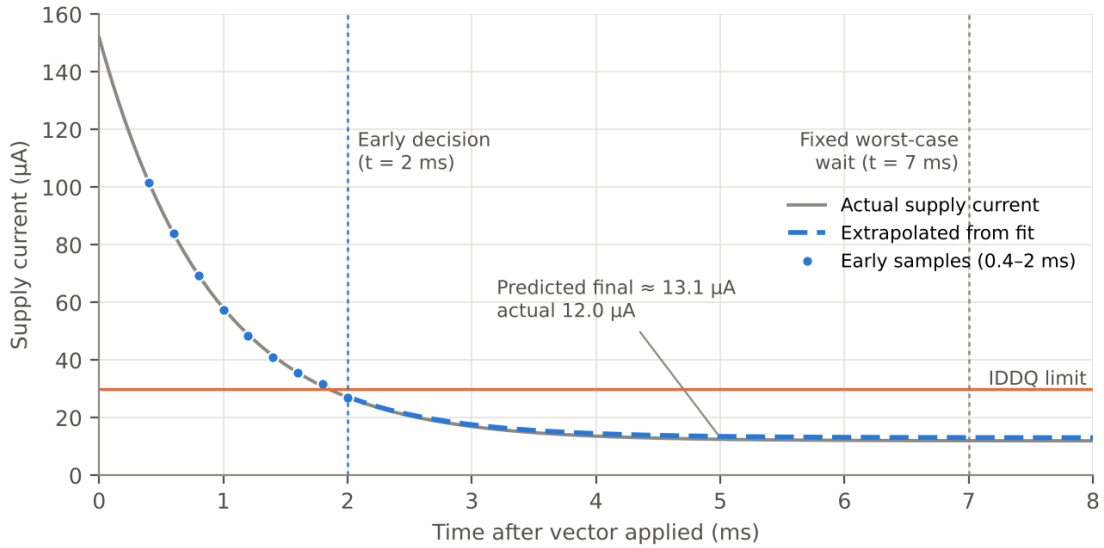


Figure 2. Early-termination prediction. Samples from 0.4–2 ms are fitted to an exponential and extrapolated; the decision is made at 2 ms instead of after the 7 ms fixed wait (synthetic example).

5.2 Estimators

Estimator	Description	When to use
Exponential curve fit	Non-linear least squares for I_{final}, A, τ (or two-term model)	Physics-based; good baseline; needs no training data
Three-point (Aitken-style) extrapolation	Closed-form estimate from three equally spaced samples: $I_{final} \approx (I_1 I_3 - I_2^2) / (I_1 + I_3 - 2I_2)$	Very fast; runs inside tester program; sensitive to noise
Regression / gradient-boosted trees	Predict I_{final} from early samples plus vector ID, mode, site, temperature, process monitors	When transients deviate from simple exponentials
Small neural network (1D-CNN / MLP)	Learns transient shape directly from sample sequence	Large datasets; complex multi-rail settling
Quantile / ensemble models	Provide prediction intervals, not just a point estimate	Required for safe decision rules

Table 3. Estimators for early termination.

5.3 Decision rule with guard band

Let $\hat{I}$ be the predicted final current and $\hat{\sigma}$ its predictive uncertainty (from residuals on validation data, a quantile model or an ensemble). With upper limit L and confidence factor k:

- **Early pass** if $\hat{I} + k \cdot \hat{\sigma} < L$
- **Early fail** if $\hat{I} - k \cdot \hat{\sigma} > L$ (optionally confirm with a full measurement before binning)
- **Otherwise** continue to the full settling time and measure conventionally

Choose k so that the expected false-pass rate on validation data is far below the product's escape budget. Because most parts sit well below the IDDQ limit, a large fraction can be decided early even with a conservative k.

```
# Early-termination IDDQ measurement (pseudocode, runs per vector)
apply_vector(v); stop_clocks()
samples = []
for t in sample_times_early:              # e.g. 0.4 ... 2.0 ms
    samples.append(measure_idd(t))
I_hat, sigma = model.predict(samples, vector=v, site=s, temp=T, monitors=m)
if I_hat + k*sigma < LIMIT[v]:
    result = PASS_EARLY                   # log samples + prediction
elif I_hat - k*sigma > LIMIT[v]:
    result = confirm_with_full_wait(v)    # never fail on prediction alone if policy requires
else:
    wait_until(T_FULL[v]); result = compare(measure_idd(), LIMIT[v])
if audit_selected(part): log(full_measurement(v))   # ground truth for monitoring
```

6. IDDQ vector reduction

IDDQ readings from different vectors are strongly correlated because most of the current is background leakage common to all vectors. Defects show up as a vector-dependent *increase* on a subset of vectors. A well-chosen subset therefore captures nearly all of the screening information.

6.1 Selection methods

- **Correlation and PCA analysis:** quantify how much independent information each vector adds; the first principal component is usually background leakage.
- **Greedy forward selection:** repeatedly add the vector that best improves prediction of the full-set screening outcome (or of the remaining vectors' readings).
- **Sparse regression (lasso):** fit the full-set decision statistic from all vectors with an L1 penalty; vectors with non-zero weights form the subset.
- **Defect-coverage weighting:** combine statistical selection with fault-coverage data so that the subset still activates the targeted bridging/defect sites.

6.2 Screening statistics that work with fewer vectors

Statistic	Idea	Benefit
Delta-IDDQ	Difference between consecutive or paired vectors removes common background	Sensitive to vector-dependent defect current
Current ratio	Ratio of max to min IDDQ across vectors	Normalizes process-driven background variation
Nearest-neighbor residual (NNR)	Compare die IDDQ to an estimate from neighboring dice on the wafer	Separates spatial process trends from true outliers
Multivariate outlier score	Mahalanobis distance or isolation-forest score on the vector set	Uses joint pattern, robust to small subsets

Table 4. IDDQ screening statistics.

```
# Greedy IDDQ vector subset selection (offline, on production data)
S = []                                  # selected vectors
target = full_set_outlier_flag          # or full-set statistic
while len(S) < max_vectors:
    # score = e.g. AUC, or recall at fixed overkill
    best = argmax_v_not_in_S  score(model_fit(X[:, S + [v]], target))
    S.append(best)
    if recall(S) >= recall_full * 0.999 and overkill(S) <= overkill_full: break
validate S on held-out lots and wafers; review with defect/fault-coverage data
```

7. Adaptive test: risk-based leakage flows

Adaptive test changes the test content per part (or per wafer/lot) based on what is already known. For leakage, the goal is to send parts that are very likely to pass through a reduced flow and keep the full flow for parts that look risky.

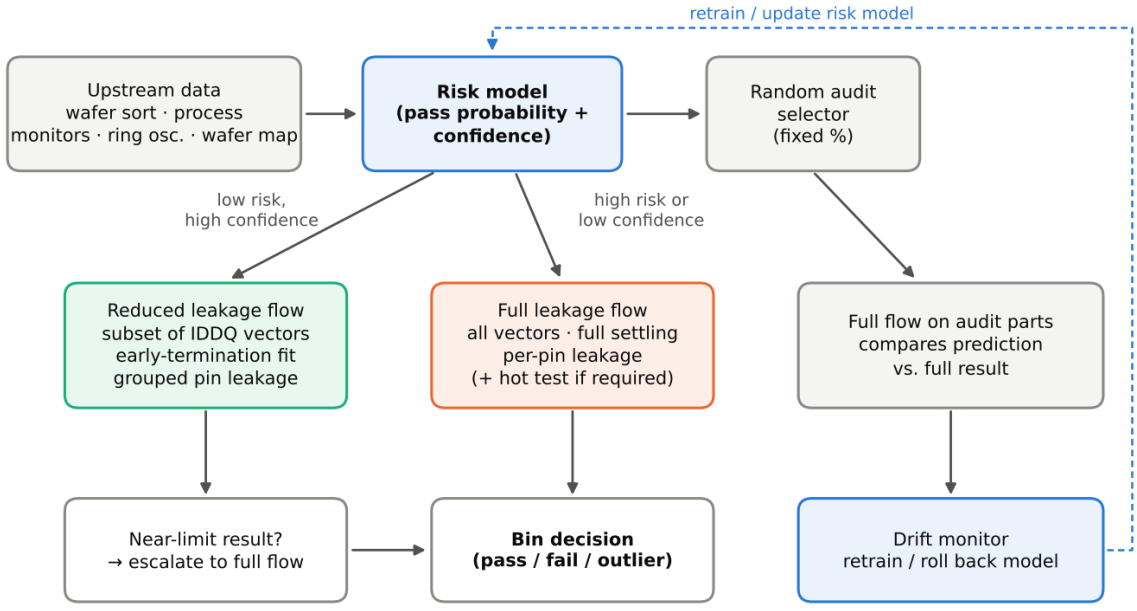


Figure 3. Risk-based adaptive leakage flow with audit sampling and drift monitoring.

7.1 Risk model

- **Inputs:** wafer-sort parametrics and leakage results (for final test), process-control monitor data, on-chip ring oscillators and leakage monitors, wafer position and neighbor results, lot history, earlier test results in the same insertion.
- **Models:** gradient-boosted trees or logistic regression for pass probability; calibrated so that predicted probabilities match observed failure rates.
- **Thresholds:** chosen by minimizing expected cost—test time saved versus the cost of an escape—subject to a hard cap on predicted escapes per million.
- **Granularity:** start at lot or wafer level (simpler qualification), then move to die level.

7.2 What the reduced flow can contain

- Reduced IDDQ vector subset (section 6)
- Early-termination measurements (section 5)
- Grouped pin leakage (section 10)
- Fewer power modes, keeping those with datasheet guarantees

8. Temperature-insertion reduction

Hot leakage testing is expensive because it needs a separate insertion and handler soak time. Leakage temperature behaviour is largely physical and predictable, so hot leakage can often be predicted from room or cold measurements plus process indicators:

$$ln\ I_{hot} = \beta_0 + \beta_1 \cdot ln\ I_{room} + \beta_2 \cdot f_{ROSC} + \beta_3 \cdot (process\ monitors) + \varepsilon$$

- Use physics-informed regression (log domain) first; add gradient-boosted trees only if residuals show structure.
- Measure the prediction interval; parts whose predicted hot leakage is close to the hot limit, or whose room-temperature behaviour is unusual (outliers in the room/hot relationship), get the hot test.
- Keep a fixed random audit fraction at hot to monitor the model.
- Temperature-activated defects may not appear at room temperature. Confirm with failure analysis data which defect types the hot test catches before reducing it.

Caution. Removing or sampling a hot insertion is the highest-value and highest-risk change in this document. For automotive and other safety-critical products it normally requires customer notification and qualification evidence. Start with optimized hot settling and vector reduction before sampling.

9. Wafer-level spatial modeling

Leakage varies smoothly across a wafer because it tracks gate length, threshold voltage and other process parameters. Spatial models exploit this:

- **Nearest-neighbor residuals:** compare each die's leakage to the median of its neighbors to separate defect outliers from process trends; this also tightens limits without extra measurements.
- **Gaussian-process (kriging) models:** predict leakage of untested or partially tested dice from measured ones, with uncertainty; useful for sampling strategies in mature, stable products.
- **Wafer signatures:** classify wafer-map patterns (edge rings, centre hot spots, scratches) to decide when a wafer needs the full flow.

10. Pin-leakage group testing

Instead of measuring each pin separately, connect a group of g pins to the same force/measure condition and measure their combined leakage. If the group passes a scaled limit, every pin in it passes. Only failing groups are split (bisected) to find the failing pin.

For P pins, group size g and a small per-pin failure probability p, the expected number of measurements is

$$N \approx P/g + (P/g) \cdot [1 - (1 - p)^g] \cdot 2 \cdot \log_2 g$$

For high-yield parts (p very small), N approaches P/g. Practical limits on g:

- Combined background leakage of g pins must stay well below the group limit, so a single leaky pin is still detectable with margin.
- Pins in a group must share compatible force voltages and must not interact (no shorts tested between them).
- The tester must support ganged PMU connections or parallel measurement.
- An ML model can choose group composition and size per lot using recent leakage distributions.

11. Data, infrastructure and real-time inference

11.1 Data

- **Datalogs:** STDF or equivalent records per part, test, site and insertion.
- **Waveform captures:** full settling transients (sampled DPS/PMU current versus time) on a characterization set and on audit parts in production. These are the training labels for early termination.
- **Upstream data:** wafer sort, process control monitors, inline metrology, wafer maps, lot genealogy.
- **Traceability:** electronic chip ID or die coordinates to join data across insertions.

11.2 Where the model runs

Option	Latency	Typical use
Inside the tester program (C/C++ or test-language code)	Microseconds	Curve fits, three-point extrapolation, small linear or tree models for early termination
Tester-attached edge server	Milliseconds	Gradient-boosted trees, neural nets, per-part risk scoring
Factory / cloud analytics	Minutes to hours	Per-lot or per-wafer flow decisions, retraining, drift monitoring

Table 5. Inference deployment options.

11.3 Model selection summary

Task	Recommended models
Early termination	Exponential/two-term fit; quantile gradient-boosted trees; small 1D-CNN for complex transients
Vector reduction	PCA, greedy forward selection, lasso / elastic net
Risk-based routing	Calibrated gradient-boosted trees, logistic regression
Hot-from-room prediction	Log-domain linear regression; GP or boosted trees on residuals
Spatial modeling	Nearest-neighbor residuals, Gaussian-process (kriging) regression
Drift detection	Population stability index, residual control charts, audit-part comparison

Table 6. Models per task.

12. Quality assurance and risk control

12.1 Metrics

Metric	Definition
Test escapes / DPPM	Defective parts shipped per million; measured via audit parts, downstream tests and field returns
Overkill	Good parts rejected; measured against full-flow results on audit parts
Decision agreement	Fraction of parts where the reduced/early decision matches the full measurement
Early-decision rate	Share of measurements decided early (time-saving driver)
Test time per part	Including site efficiency and handler index time
Correlation	Correlation of predicted vs. measured leakage, per test, per site, per tester

Table 7. Quality and efficiency metrics.

12.2 Safeguards

- **Shadow mode:** run the AI decision alongside the full test for a statistically meaningful number of lots before acting on it.
- **Held-out validation:** validate on lots, wafers and testers not used in training.
- **Guard bands:** only decide early or skip when the prediction is well inside limits; borderline parts always get the full measurement.
- **Audit sampling:** a fixed random fraction of parts gets the full flow permanently.
- **Drift monitoring and rollback:** alarms on input distribution shifts (new fab, mask, process change, tester hardware swap) and on audit disagreement; automatic fallback to the full flow.

- **Change control:** treat model updates like test-program changes, with versioning, review and release.

12.3 Automotive and safety-critical products

Automotive programs typically follow AEC-Q100 qualification and use Part Average Testing (AEC-Q001) and related outlier screens, with zero-defect targets. Settling optimization and statistically validated vector reduction are usually easier to justify than skipping measurements. Any reduction of test content should be agreed with the quality organization and, where required, notified to customers.

13. Implementation roadmap

Phase	Duration (typical)	Activities	Exit criteria
0. Baseline	1–2 weeks	Profile leakage test time per test, vector, mode, pin group and insertion	Time breakdown and target savings agreed
1. Characterization	2–4 weeks	Capture full settling waveforms across corners, sites and testers	Settling database covering process/temperature spread
2. Quick wins	2–6 weeks	Per-vector waits, fixed ranging, integration tuning, group pin leakage, multi-site	Released program with measured time saving and unchanged yield/quality
3. Early termination	6–10 weeks	Build estimators, decision rule and guard bands; shadow mode	Decision agreement and false-pass rate within target on held-out lots
4. Vector reduction	4–8 weeks	Subset selection on production data; defect-coverage review	Recall and overkill equal to full set on held-out data
5. Adaptive flows	8–16 weeks	Risk model, routing, audit sampling, drift monitoring	Quality board approval; production release with rollback
6. Temperature reduction	Product-dependent	Hot-from-room models; FA review; customer approval if needed	Evidence package accepted; audit plan in place

Table 8. Phased roadmap.

14. Business case

Test cost scales with tester time. A simple estimate of annual savings is

$$Savings/year \approx \Delta T_{part} \times Volume_{year} \times C_{tester} / (N_{sites} \times site\ efficiency)$$

where ΔT_{part} is the time saved per part (seconds), C_{tester} is the fully loaded tester cost per second, and N_{sites} is the number of parallel sites. Include the one-time cost of characterization, model development and qualification, and the ongoing cost of audit sampling and monitoring. Time savings can also be taken as capacity: fewer testers needed for the same volume.

15. Glossary

Term	Meaning
ATE	Automatic test equipment
DPS / PMU	Device power supply / parametric measurement unit on the tester
DPPM	Defective parts per million
GR&R	Gauge repeatability and reproducibility
IDDQ	Quiescent supply current measured with clocks stopped after a test vector
I_{IL} / I_{IH} / I_{OZ}	Input low / input high / tri-state output leakage current
ISB	Standby supply current
NNR	Nearest-neighbor residual, a spatial outlier statistic
PAT	Part Average Testing, statistical outlier screening (AEC-Q001)
ROSC	Ring oscillator used as an on-chip process monitor
STDF	Standard Test Data Format for ATE datalogs
τ	Settling time constant of the supply current transient

16. Further reading

[1] W. R. Daasch, J. McNames, D. Bockelman and K. Cota, “Variance reduction using wafer patterns in IDDQ data,” *Proc. IEEE International Test Conference (ITC)*, 2000.

[2] S. Sabade and D. M. H. Walker, “Improved wafer-level spatial analysis for IDDQ limit setting,” *Proc. IEEE International Test Conference (ITC)*, 2001.

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